

SSH Client

Your Laptop

SSH Server

Linux VM

1. Client initiates SSH connection



2. Server sends its public identity

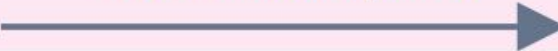


3. Challenge encrypted with client's public key

Only the client's private key can decrypt it



4. Authentication passed



5. Session key created

All terminal traffic is now encrypted

Key	Purpose
Public Key	Encrypt/verify identity
Private Key	Decrypt authentication challenge
Session Key	Encrypt all SSH traffic

Every command is encrypted by Session Key:

```
ls
```

```
cd /var
```

```
sudo apt update
```

Even if someone captures the packets, they'll only see encrypted data.

1. Client → Server: "I want to connect."
2. Server → Client: Sends its host public key (proves server identity).
3. Server: Takes the client's public key (stored in `~/.ssh/authorized_keys`), creates a random challenge, and encrypts it.
4. Client: Uses its private key to decrypt the challenge and sends the correct response.
5. Server: Verifies the response and grants access.

At Stage 2

Are you sure you want to continue connecting (yes/no)?

When you type yes, the server's fingerprint is saved in:

`~/.ssh/known_hosts`

Note : This protects against Man-in-the-Middle attacks.

Commands

At Server side

```
$sudo apt update
```

```
$sudo apt install openssh-server
```

```
$sudo systemctl status ssh
```

```
$ip r
```

At Client Side

```
$ssh -V
```

```
$sudo apt install openssh-client
```

```
$ssh username@server-ip
```

Telnet

Plain text

Insecure

Password visible

Legacy

SSH

Encrypted

Secure

Password protected

Industry standard